

## GUAIZIHU, China

WMO: 523780

Lat: 41.367N

Lon: 102.367E

Elev: 960

StdP: 90.31

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -21.8      | -19.5 | -32.4                         | 0.2 | -11.6 | -30.3 | 0.3 | -10.7 | 12.4                  | -7.6 | 11.2 | -7.2 | 1.8                   | 90   | 0.661 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.9                   | 38.0            | 17.1 | 36.5 | 16.5 | 35.0 | 16.0 | 19.8                | 31.1 | 18.7 | 29.9 | 17.8 | 29.3 | 5.5                  | 90   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 16.8                            | 13.5 | 22.7 | 15.3 | 12.2 | 22.9 | 13.6 | 10.9 | 23.4 | 61.4          | 31.0 | 57.4 | 29.9 | 54.3 | 29.3 | 27.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |    |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |    |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   | DB   | WB |
| (a)               | (b)  | (c)  | (d)                        | (e)   | (f)                | (g) | (h)                                                | (i)   | (j)        | (k)   | (l)        | (m)   | (n)        | (o)   |      |    |
| 13.2              | 11.6 | 10.3 |                            | -24.4 | 40.5               | 2.5 | 1.6                                                | -26.2 | 41.7       | -27.7 | 42.6       | -29.1 | 43.5       | -31.0 | 44.7 |    |
|                   |      |      |                            | -24.8 | 21.2               | 2.4 | 2.0                                                | -26.5 | 22.7       | -27.9 | 23.9       | -29.3 | 25.0       | -31.0 | 26.5 |    |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 10.5  | -10.1 | -4.5  | 3.7   | 13.3  | 19.7  | 25.7  | 28.3  | 26.1  | 19.7  | 10.5  | 0.7   | -7.9 | (6)  |
| (7)  |                                                                   | DBStd       | 13.99 | 4.52  | 5.04  | 5.68  | 5.13  | 4.35  | 3.31  | 3.20  | 3.25  | 4.20  | 4.69  | 4.99  | 4.85 | (7)  |
| (8)  |                                                                   | HDD10.0     | 2147  | 623   | 407   | 206   | 25    | 1     | 0     | 0     | 1     | 51    | 279   | 556   |      | (8)  |
| (9)  |                                                                   | HDD18.3     | 3794  | 881   | 640   | 454   | 162   | 36    | 1     | 0     | 1     | 33    | 244   | 528   | 814  | (9)  |
| (10) |                                                                   | CDD10.0     | 2333  | 0     | 0     | 10    | 125   | 302   | 470   | 568   | 499   | 291   | 67    | 1     | 0    | (10) |
| (11) |                                                                   | CDD18.3     | 939   | 0     | 0     | 0     | 12    | 79    | 221   | 309   | 242   | 73    | 2     | 0     | 0    | (11) |
| (12) |                                                                   | CDH23.3     | 11893 | 0     | 0     | 1     | 149   | 962   | 2842  | 4264  | 2933  | 718   | 24    | 0     | 0    | (12) |
| (13) |                                                                   | CDH26.7     | 6112  | 0     | 0     | 0     | 36    | 371   | 1469  | 2473  | 1509  | 251   | 2     | 0     | 0    | (13) |
| (14) | Wind                                                              | WSAvg       | 4.8   | 3.9   | 4.3   | 5.2   | 5.7   | 5.6   | 5.3   | 5.1   | 4.8   | 4.5   | 4.4   | 4.6   | 4.2  | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 34    | 0     | 0     | 1     | 1     | 4     | 4     | 11    | 9     | 4     | 3     | 0     | 0    | (15) |
| (16) |                                                                   | PrecMax     | 78    | 2     | 1     | 6     | 14    | 30    | 36    | 45    | 28    | 18    | 22    | 4     | 4    | (16) |
| (17) |                                                                   | PrecMin     | 12    | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0    | (17) |
| (18) |                                                                   | PrecStd     | 16    | 1     | 0     | 2     | 3     | 6     | 7     | 10    | 7     | 5     | 6     | 1     | 1    | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB    | 3.9   | 10.6  | 21.5  | 30.2  | 35.2  | 38.5  | 40.8  | 38.7  | 34.4  | 26.1  | 16.0  | 5.9  | (19) |
| (20) |                                                                   | MCWB        | -1.7  | 2.1   | 7.4   | 11.8  | 13.8  | 16.1  | 18.8  | 17.8  | 15.3  | 11.2  | 5.7   | -0.7  |      | (20) |
| (21) |                                                                   | 2%          | DB    | 0.9   | 7.8   | 18.3  | 27.2  | 32.7  | 36.5  | 38.8  | 36.6  | 31.5  | 23.2  | 13.2  | 3.1  | (21) |
| (22) |                                                                   | MCWB        | -3.8  | 0.3   | 5.9   | 10.3  | 13.1  | 15.5  | 17.6  | 17.0  | 14.3  | 9.7   | 4.1   | -2.3  |      | (22) |
| (23) |                                                                   | 5%          | DB    | -1.1  | 5.8   | 15.6  | 24.6  | 30.6  | 35.1  | 37.3  | 34.9  | 29.5  | 20.9  | 10.9  | 1.1  | (23) |
| (24) |                                                                   | MCWB        | -5.0  | -0.8  | 4.5   | 9.3   | 12.3  | 15.0  | 17.1  | 16.4  | 13.5  | 8.6   | 2.9   | -3.6  |      | (24) |
| (25) |                                                                   | 10%         | DB    | -2.9  | 3.8   | 12.9  | 22.3  | 28.6  | 33.5  | 35.7  | 33.4  | 27.5  | 18.6  | 8.7   | -0.7 | (25) |
| (26) |                                                                   | MCWB        | -6.3  | -2.0  | 3.2   | 8.3   | 11.6  | 14.5  | 16.7  | 16.1  | 12.8  | 7.5   | 1.7   | -4.7  |      | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB    | -1.7  | 2.5   | 7.7   | 12.5  | 15.4  | 18.5  | 22.1  | 21.3  | 18.2  | 12.3  | 5.9   | -0.5 | (27) |
| (28) |                                                                   | MCDWB       | 3.5   | 9.4   | 19.9  | 27.1  | 30.1  | 27.1  | 32.0  | 29.3  | 26.5  | 22.3  | 14.4  | 5.6   |      | (28) |
| (29) |                                                                   | 2%          | WB    | -3.5  | 0.7   | 6.2   | 11.0  | 14.0  | 17.4  | 20.3  | 19.7  | 16.3  | 10.3  | 4.4   | -2.2 | (29) |
| (30) |                                                                   | MCDWB       | 0.7   | 7.2   | 16.8  | 25.3  | 30.0  | 28.4  | 32.8  | 30.5  | 26.0  | 21.5  | 12.7  | 2.7   |      | (30) |
| (31) |                                                                   | 5%          | WB    | -4.9  | -0.6  | 4.9   | 9.8   | 13.0  | 16.6  | 19.1  | 18.7  | 14.9  | 9.1   | 3.1   | -3.4 | (31) |
| (32) |                                                                   | MCDWB       | -1.3  | 5.5   | 14.7  | 23.4  | 28.2  | 29.9  | 30.9  | 29.3  | 25.7  | 19.7  | 10.7  | 1.0   |      | (32) |
| (33) |                                                                   | 10%         | WB    | -6.3  | -2.0  | 3.5   | 8.6   | 12.1  | 15.8  | 18.2  | 17.6  | 13.7  | 7.8   | 1.7   | -4.6 | (33) |
| (34) |                                                                   | MCDWB       | -3.0  | 3.6   | 12.5  | 21.7  | 26.9  | 30.2  | 30.1  | 28.9  | 25.1  | 17.7  | 8.5   | -0.9  |      | (34) |
| (35) | Mean Daily Temperature Range                                      | MDBR        | 11.6  | 13.2  | 13.6  | 12.9  | 12.9  | 12.4  | 11.9  | 12.0  | 12.3  | 12.7  | 11.5  | 10.6  |      | (35) |
| (36) |                                                                   | 5% DB       | MCDBR | 12.4  | 14.8  | 16.1  | 15.3  | 15.3  | 14.2  | 13.5  | 13.6  | 14.0  | 14.8  | 13.4  | 10.8 | (36) |
| (37) |                                                                   | MCWBR       | 9.1   | 9.6   | 9.0   | 7.3   | 6.8   | 5.2   | 4.2   | 4.5   | 5.3   | 7.5   | 7.8   | 7.5   |      | (37) |
| (38) |                                                                   | 5% WB       | MCDBR | 12.3  | 14.3  | 14.9  | 14.0  | 13.6  | 11.7  | 11.0  | 11.6  | 12.2  | 14.0  | 13.3  | 10.7 | (38) |
| (39) |                                                                   | MCWBR       | 9.0   | 9.3   | 8.5   | 6.9   | 6.3   | 4.1   | 3.8   | 4.2   | 4.7   | 7.2   | 7.9   | 7.6   |      | (39) |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.317 | 0.372 | 0.417 | 0.464 | 0.479 | 0.494 | 0.486 | 0.466 | 0.431 | 0.381 | 0.341 | 0.318 |      | (40) |
| (41) |                                                                   | taud        | 2.139 | 2.012 | 1.914 | 1.842 | 1.863 | 1.892 | 1.963 | 2.001 | 2.027 | 2.082 | 2.122 | 2.126 |      | (41) |
| (42) |                                                                   | Ebn at Noon | 807   | 807   | 815   | 806   | 804   | 792   | 794   | 797   | 794   | 788   | 766   | 765   |      | (42) |
| (43) |                                                                   | Edh at Noon | 103   | 138   | 172   | 199   | 199   | 194   | 179   | 168   | 152   | 127   | 105   | 97    |      | (43) |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 2.58  | 3.66  | 4.97  | 6.25  | 7.13  | 7.16  | 6.65  | 6.17  | 5.25  | 4.15  | 2.88  | 2.30  |      | (44) |
| (45) |                                                                   | RadStd      | 0.18  | 0.19  | 0.38  | 0.28  | 0.33  | 0.40  | 0.35  | 0.32  | 0.29  | 0.13  | 0.14  | 0.19  |      | (45) |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only           | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (2 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +82     | +44     |

Nomenclature: See separate page